Precision IoT

Complete guide to monitoring your sensor network, managing devices, and reviewing intelligent ML-generated alerts from the mobile app.

Mobile App Guide | ML Alerts | Live Streaming | Last updated: April 2026

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Getting Started

Follow these steps the first time you open the app, whether you are signing in with Google or creating a non-Gmail email/password account.

Signing In

1 Open the App

Launch Precision IoT on your mobile device. The login screen appears with two sign-in options.

2 **Sign In with Google (recommended)**

Tap **Sign in with Google**. Choose your Google account from the picker. Your name and photo load automatically.

3 **— or — Sign In with Email & Password**

Tap **Sign in with Email**, enter your registered email and password, then tap **Sign In**. This path is for accounts that have already been created and approved by an administrator.

4 **Automatic Redirect to Dashboard**

On success you are taken straight to the Dashboard. Your profile is synced in the background.

Creating a New Email/Password Account

1 **Open the Sign-Up Popup**

On the login screen, tap **Sign in with Email**, then choose the **Sign Up** option in the popup.

2 **Enter Required Details**

Provide a **non-Gmail email address**, your display name, and a password with at least 6 characters. Gmail addresses should use the Google sign-in path instead.

3 **Submit and Wait for Confirmation**

Tap **Submit**. The app creates your account and sends the new credentials to the backend. A success message appears only after the backend acknowledges receipt.

4 **Read the Success Message**

When registration completes, the app shows **New user created successfully, wait for admin approval.** At this stage the account exists, but app access is still blocked.

5 **Wait for Administrator Approval**

An administrator must unblock your account in the admin portal before your first email/password login will succeed.

6 **Sign In After Approval**

Once approved, return to **Sign in with Email** and log in with the same email and password to reach the Dashboard.

ℹ️ **New user?**

Your first Google Sign-In automatically creates an active account. If you want an email/password account, use the non-Gmail sign-up flow above.

⚠️ **Pending approval or blocked account**

If your account is waiting for administrator approval, or an administrator has blocked it later, the Dashboard will remain inaccessible until the account is unblocked.

📊 Dashboard Overview

The Dashboard is the central hub of the app. Two tabs sit at the top; a profile icon is in the header.

🖥️ Devices Tab

Lists all your connected Raspberry Pi sensor devices. Claim new devices, view live status, and manage sensors or cameras.

🚨 Alerts Tab

Displays the real-time ML Intelligence Feed. Review, rate, and manage automated detection alerts.

📡 Network Health

At the bottom of the Devices tab — global latency meter and live bar chart of connectivity across your device fleet.

🔔 Push Notifications

Critical alerts are pushed to your device instantly, even when the app is in the background.

🖥️ Managing Devices

Adding (Claiming) a Device

1 **Tap "+ Add Device"**

Button at the top of the Devices tab. Opens the list of unclaimed devices on the network.

2 **Select a Device**

Tap the device you want to claim. Only devices not currently owned by another user appear here.

3 **Confirm Claim**

Confirm in the dialog. The device is now yours and appears in your list.

Reading a Device Card

Field	What it means
Device Name	Human-friendly label you assigned (e.g., "Front Gate Camera")
Device ID	Unique system identifier (auto-generated)
Status	ACTIVE = online OFFLINE = unreachable
Last Pulse	Time since the device last contacted the server (e.g., "2 min ago")
Uptime	Availability percentage since device was added (e.g., "99.7%")

Device Actions

SENSORS Open the sensor management panel for this device.

CAMERA Launch the live HLS video stream in the in-app player.

RENAME Change the display label for easier identification.

REMOVE Un-claim this device from your account. It can be re-claimed later; no hardware is deleted.

Network Health Visualization

Metric	Description
Global Latency	Average response time (ms) across all your devices. Under 200 ms is ideal.
Latency Chart	Bar chart showing latency per poll cycle. Blue bars = recent active periods.
Connected Devices	Number of devices currently reporting a heartbeat.

Sensor Control

Tap **SENSORS** on any device card to open that device's sensor management panel.

Available Operations

Action	How To	Effect
View sensors	Open the SENSORS panel	Lists every sensor with its label, GPIO pin, and ON/OFF state
Toggle ON / OFF	Tap the ON or OFF button next to a sensor	Sends a remote command to enable or disable that sensor on the device
Add sensor	Enter label and GPIO pin number, tap Add	Registers a new sensor; the device picks it up on next poll
Delete sensor	Tap the delete icon on a sensor row	Removes sensor from the system (requires confirmation)

ℹ️ **Data refresh rate**

Live temperature and humidity readings poll every 15 seconds (DHT11 sensor). A toggle may take up to one cycle to propagate.

⚠️ **Access required**

You can only control sensors on devices you own or have been granted controller-level access to by an administrator.

📹 Live Camera Streaming

Each Raspberry Pi device can stream live video directly into the app using HLS (HTTP Live Streaming).

Starting a Live Stream

1. **Tap CAMERA on a Device Card**
   The in-app HLS video player opens full-screen automatically.
2. **Stream Buffers and Plays**
   A loading indicator appears while the stream buffers. Playback begins automatically — no extra tap needed.
3. **Continuous Loop**
   The stream loops so you always see the latest footage without manually restarting.
4. **Close the Player**
   Tap the ✕ close button (top-right) to exit and return to the Devices tab.

What the Player Shows

Item	Details
Device Label	Shown at top (e.g., "📹 Front Gate — Live Stream")
Loading Spinner	Visible while stream buffers; disappears when live
Status Messages	Connecting..., Stream ready, or error text shown inline

Item	Details
Auto Retry	Player reconnects automatically on transient network failures

> ⚠️ **Device must be ACTIVE**
> The stream will not load if the device shows OFFLINE. Verify the Raspberry Pi is powered on and connected to the internet.

🚨 ML Alerts & Ratings

The **Alerts tab** shows a live feed of events detected by the machine-learning model running on your sensor devices.

Alert Retention Settings

Choose how long alerts are kept before automatic deletion. Tap your preferred period at the top of the Alerts tab:

7 days | **15 days** | **30 days** | **Never**

Your selection saves automatically. "30 days" is the default. Use *Never* only for critical records.

Reading an Alert Card

Component	Description
Risk Badge	CRITICAL HIGH MEDIUM LOW
Device Name	Which sensor device triggered the alert
Detected Objects	What the ML model found (e.g., "person", "vehicle", "fire")
Confidence	Model certainty as a percentage (e.g., "92.0% CONFIDENCE")
Preview Image	Thumbnail captured at the moment of detection (tap to enlarge)

Component	Description
Timestamp	Relative time since alert (e.g., "3 min ago")
Accuracy Status	✓ **Accurate** or ✗ **Inaccurate** once rated
Rating	⭐ your 1–10 score (shown after rating)

Risk Level Colour Guide

🔴 CRITICAL — Immediate action required

🟠 HIGH — Prompt attention needed

🟡 MEDIUM — Review when possible

🔵 LOW — Informational only

Rating an Alert (Feedback)

1. **Tap an Alert Card**
   Opens the Rating dialog for that alert.

2. **Mark Accuracy**
   Tap **Accurate** if the detection was correct, or **Inaccurate** for a false positive.

3. **Select a Score (1–10)**
   Tap a number to rate the usefulness of the detection. 10 = very useful.

4. **Tap OK / Submit**
   Saved instantly. Card background updates to green (accurate) or red (inaccurate).

> ✅ **Why rate alerts?**
> Every rating is used to retrain the ML pipeline. Consistent feedback leads to fewer false alarms within days.

Viewing Full Alert Details

Tap **VIEW DETAILS** on any alert card to open a modal with:

- Full-resolution capture image from the moment of detection
- Complete detection metadata: detected classes, confidence, bounding boxes
- Original risk level assigned by the model

🔔 Push Notifications

The app pushes a notification to your device the moment a new ML alert is generated, even when the app is not open.

Enabling Notifications

1. **Allow on First Launch**
   The app requests notification permission on sign-in. Tap **Allow**.

2. **Re-enable if Denied**
   Go to phone *Settings → Apps → Precision IoT → Notifications* and turn them on.

Notification Format

Part	Example
Title	🚨 CRITICAL
Body	person, vehicle — raspberrypi-front-gate
Sound & Vibration	Enabled by default for all risk levels
Tap action	Opens the app on the Alerts tab

> ℹ️ **No duplicates**
> Each alert triggers exactly one notification, even if the app is reopened multiple times.

Profile & Sign Out

Viewing Your Profile

Tap the profile icon in the top-right of the Dashboard to open your profile panel. It shows your display name, profile photo, registered email, and sign-in method.

Signing Out

1. **Open Profile Panel**
   Tap the profile icon in the top-right corner.

2. **Tap Sign Out**
   Confirm in the dialog. Session is cleared and the app returns to the Login screen.

⚠️ **Your data is safe**

Signing out does not delete devices or alert history. Everything is restored on next sign-in.

Blocked Account Access

If your account is pending administrator approval or has been blocked, the app prevents Dashboard access until an administrator restores access.

What you see	What it means
Account pending activation	Your email/password account was created successfully and is still waiting for administrator approval
Access Denied screen	A system administrator has blocked an account that was previously active
Block reason text	The reason provided by the administrator, if one was supplied

What you see	What it means
Sign Out button	Only available action — clears your local session

🚫 **Cannot access app while approval is pending or while blocked**

You cannot view devices, sensors, or alerts until an administrator approves or restores your access. Contact your system administrator if approval is delayed.

🔧 Troubleshooting

Sign-in fails or loops

- Ensure a stable internet connection.
- Use the correct Google account (the one with access).
- Close and reopen the app, then try again.
- For email/password: check caps lock and trailing spaces.

Device shows OFFLINE

- Confirm the Raspberry Pi is powered on.
- Verify Wi-Fi / internet connectivity on the device.
- Restart the device and wait 30 seconds to reconnect.
- If persistent, contact your system administrator.

Camera stream won't load

- Confirm the device status is **ACTIVE**.
- Check your own internet connection.
- Close and reopen the player — it retries automatically.
- Verify no firewall rule is blocking the stream URL.

Sensor toggle has no effect

- Confirm the device is online.
- Wait one poll cycle (~15 seconds) for state to propagate.
- Restart the device if state remains stuck.
- Make sure you have controller-level access to this device.

Push notifications not arriving

- Go to phone *Settings* → *Notifications* → *Precision IoT* and allow notifications.
- Sign out and back in to re-register your push token.
- Disable Battery Saver mode that may block background activity.

Alerts feed is empty

- Check your **Alert Retention** setting — old alerts may have been auto-deleted.
- Verify at least one device is online and the ML service is running.
- Pull down to refresh the Alerts tab.

❓ Need Help?

If you encounter an issue not covered in this guide, contact your system administrator or the Precision IoT support team.

For technical documentation, API references, and admin configuration guides, refer to the full system documentation index.

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App User Guide — April 2026